

- ① Set angular velocity to zero when it reverses direction:

$$0 = A - Bt^2$$

$$t = \sqrt{\frac{A}{B}} = \sqrt{\frac{18}{0.50}} = \sqrt{36} = 6.0$$

Answer: 6.0

- ② The disk reverses at $t = 6.0$ s

Angular displacement:

$$\Delta\theta = \int_0^6 (A - Bt^2) dt$$

$$\Delta\theta = At - \left(\frac{B}{3}\right)t^3$$

$$\Delta\theta = 18(6.0) - \left(\frac{0.50}{3}\right)(6.0)^3$$

$$\Delta\theta = 108 - 36 = 72$$

Answer: 72